

Seungheon (Hoon) Yang

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Engineering

Sep 2021 – Mar 2027

Cumulative GPA: 3.71/4.00

Honors: IEEE-HKN Eta Kappa Nu Honor Society Inductee; Dean's Honor List, 9 of 10 quarters

Certifications: SQLD, AWS Certified Machine Learning Associate, AWS Certified AI Practitioner, AWS Certified Cloud Practitioner

Relevant Coursework: Introduction to Machine Learning, Data Structures and Algorithms, Computer Architecture, System Software, Computer Networks, Advanced C Programming

PROFESSIONAL EXPERIENCE

Amazon Web Services – ProServe A2C

Jun 2026 – Sep 2026

AI/ML Intern

Seoul, KR

Agentic Readiness Screener | *Individual Project*

- Built an internal platform that maps customer requirements and database schemas to actionable feasibility reports, accelerating a 1-2 week ad-hoc process that 7 of 10 delivery consultants identified as a recurring pain point
- Designed an Agents-as-Tools human-in-the-loop workflow and deployed it via FastAPI on Amazon ECS Fargate, achieving 85% reproducibility and 100% loop validity
- Architected a knowledge graph retrieval service using Amazon Neptune and OpenSearch to identify missing schema concepts, reaching 78.7% Recall@6 on a leave-one-out evaluation

Knowledge-Sharing Hub | *Team Project*

- Engineered an internal video knowledge platform that converts session recordings into searchable reports and a session-specific chatbot, designing an E2E pipeline with Strands Agents orchestration on Amazon Bedrock AgentCore Runtime and cutting processing time for a 30-minute video by 63%
- Optimized a multimodal slide-search engine using Cohere Embed v4 with hybrid kNN vector + BM25 OCR retrieval, improving nDCG@10 from 0.39 to 0.88 and top-1 accuracy from 14% to 100%

AICPS Lab, UC Irvine

Sep 2025 – Jun 2026

Undergraduate Research Assistant

Irvine, CA

- Developed a vision-language navigation evaluation pipeline in the OpenUAV simulator for trajectory analysis, metric extraction, and video generation, automating 28 evaluation runs
- Reimplemented the TypeFly baseline to run controlled comparative experiments across 7 task-based navigation scenarios
- Performed monocular 3D reconstruction from physical UAV image sequences using MonST3R and generated visualizations in Blender

PROJECTS

ZotTARS

Sep 2025 – Jun 2026

Software Subteam

- Architected a modular FastAPI backend by mapping ROS-style node-based decoupling to the STT-LLM-TTS pipeline, enabling independent services for a real-time voice AI chatbot system
- Implemented a low-latency real-time control environment integrating PS5, GUI, and Arduino into a unified system
- Integrated YOLO-based object detection into a robotic assistant, enabling vision-grounded LLM responses and control decisions

Speed Violation Tracker

Mar 2026 – Jun 2026

- Developed an end-to-end computer vision pipeline for real-time vehicle detection, tracking, and speed-violation capture from video and livestream using YOLO and OpenCV, improving display FPS by 3.4× via decoupled inference and rendering threads
- Estimated vehicle speeds via perspective transformation, mapping user-defined trapezoidal ROIs to real-world coordinates with manual distance calibration
- Built a Flask web application with video upload, interactive browser-based ROI drawing, and Server-Sent Events for real-time log streaming of detection results

Autonomous Spider Bot

Jan 2026 – Jun 2026

- Owned sensing and compute integration for a multi-legged robot, configuring LiDAR and IMU sensors on an NVIDIA Jetson Orin Nano with secure remote access via Tailscale
- Engineered a Proximal Policy Optimization algorithm for autonomous robot navigation and validated it in PyBullet simulations

SKILLS

Programming: Python, C/C++, Java, JavaScript, SQL, HTML/CSS

AI/ML & CV: PyTorch, OpenCV, Ultralytics, LLM Agents, RAG, Reinforcement Learning

Cloud & Tools: Amazon Bedrock, AgentCore, Lambda, DynamoDB, S3, OpenSearch, ECS Fargate, Neptune, FastAPI, Linux, Git

Spoken Languages: Native in Korean, Fluent in English